

Rapid Detection of Extended-Spectrum- β -Lactamase-Producing Enterobacteriaceae



Enterobacterial strains producing clavulanic-acid-inhibited extended-spectrum β -lactamases (ESBLs) are increasingly reported worldwide. Conventional detection of ESBL production remains time-consuming (24 to 48 h). Therefore, the ESBL NDP (Nordmann/Dortet/Poirel) test was developed for a rapid identification of ESBLs in Enterobacteriaceae. This biochemical test was based on the in vitro detection of a cephalosporin (cefotaxime) hydrolysis that is inhibited by tazobactam addition. The ESBL activity was evidenced by a color change (red to yellow) of a pH indicator (red phenol) due to carboxyl-acid formation resulting from cefotaxime hydrolysis that was reversed by addition of tazobactam (positive test). The ESBL NDP test was applied to cultured strains (215 ESBL producers and 40 ESBL nonproducers). Its sensitivity and specificity were 92.6% and 100%, respectively. Its sensitivity (100%) was excellent for detection of CTX-M producers. A few ESBL producers (n = 16) that remained susceptible to cefotaxime were not detected. The test was also evaluated on spiked blood cultures and showed excellent sensitivity and specificity (100% for both). The test was rapid (less than 1 h) and cost-effective. It can be implemented in any health care facility and is well adapted for infection control purposes in particular.